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SnailLoad: Exploiting Remote Network Latency Measurements without JavaScript

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Inferring user activities on a computer from network traffic is a well-studied attack vector. Previous work has shown that they can infer websites visited, videos watched, and even user actions within specific applications. However, all of these attacks require a scenario where the attacker can observe the (possibly encrypted) network traffic, e.g., through a person-in-the-middle (PITM) attack or sitting in physical proximity to monitor WiFi packets.

In this paper, we present SnailLoad, a new side-channel attack where the victim loads an asset, e.g., a file or an image, from an attacker-controlled server, exploiting the victim's network latency as a side channel tied to activities on the victim system, e.g., watching videos or websites. SnailLoad requires no JavaScript, no form of code execution on the victim system, and no user interaction but only a constant exchange of network packets, e.g., a network connection in the background. SnailLoad measures the latency to the victim system and infers the network activity on the victim system from the latency variations. We demonstrate SnailLoad in a non-PITM video-fingerprinting attack, where we use a single SnailLoad trace to infer what video a victim user is watching momentarily. For our evaluation, we focused on a set of 10 YouTube videos the victim watches, and show that SnailLoad reaches classification F1 scores of up to 98%. We also evaluated SnailLoad in an open-world top 100 website fingerprinting attack, resulting in an F1 score of 62.8%.

This shows that numerous prior works, based on network traffic observations in PITM attack scenarios, could potentially be lifted to non-PITM remote attack scenarios.

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